

Module 10: Confidence Intervals for Proportions

Introduction

Often, we will not know the true population parameter, so we will need to estimate it. The sample proportion is used as a **point estimate** of the population proportion.

- A **point estimate** is a single value based on representative sample data that is a plausible estimate of the population parameter.
- For proportions this estimate is $\hat{p}$.
- Mean: $\mu_{\hat{p}} = p$
- Standard Deviation: $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$
- If p is not known and the population is large enough front he 10% condition, then we can use the standard error to determine variability for the point estimate.

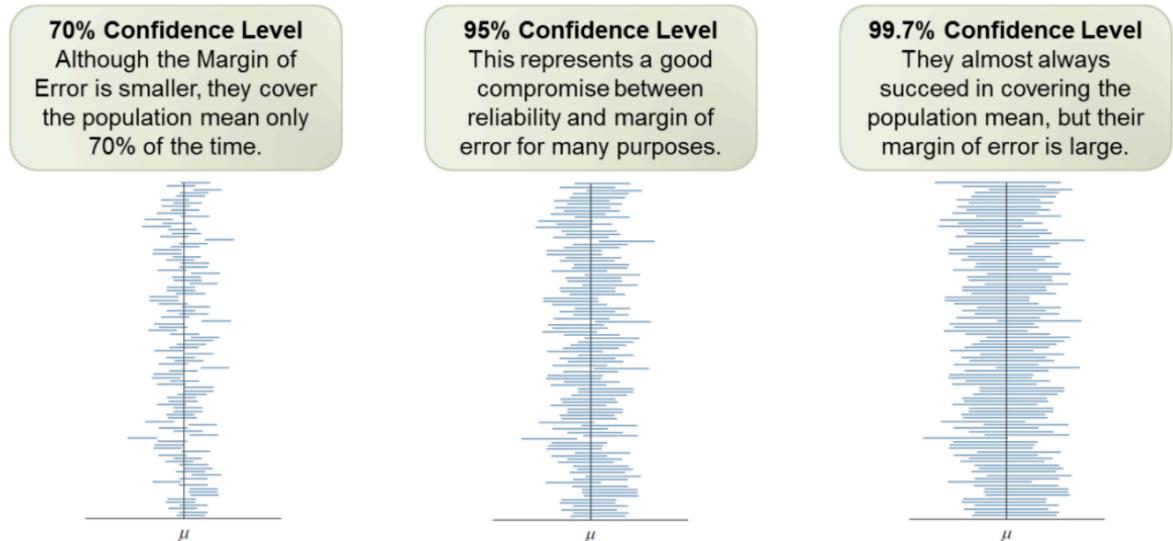
- $S.E. = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

- We create confidence intervals because, as previously seen, sample proportions vary from sample to sample.
- Confidence intervals take sampling variability into account and convey information on point estimate accuracy.
- A confidence interval is calculated using the point estimate and the margin of error.

Margin of Error

- The **margin of error (ME)** is what determines the width of the interval. A confidence interval will have a width of twice the margin of error. It is calculated using the **standard error** and the z –**critical value** for the confidence level.

- If we want to be more confident that our interval contains the true value, we must increase the margin of error. There is a trade-off. We would rather have a higher level of confidence than a lower level, but we would also rather have a smaller margin of error than a larger one.



The Margin of Error Depends on the Following Factors

1. **Population standard deviation:** the normal distribution can be used to find the confidence level when the population standard deviation is known.
2. **Confidence level:** if a large number of intervals are constructed, the confidence level represents the expected percentage of intervals that would contain the parameter.
3. **Sample size:** the margin of error decreases as the sample size increases.

Summary of Conditions for C.I.

1. Random: SRS or randomized sampling design
2. Independent: Population needs to be at least 10n
3. Normal: $np \geq 10$ and $n(1 - p) \geq 10$

****The higher the confidence, the wider the interval!!****

Interpretation: We can be 99% confident that the population proportion of ____ falls between ____ and ____.

Constructing a Confidence Interval for p

Statistic $\pm$ (critical value) * (standard deviation of statistic)

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

How to determine the critical value, z^* ?

You find the z-score associated with that given area. For example, if you want an 80% C.I., then you find the z-score associated with $\frac{(1-.80)}{2} = \frac{.20}{2} = .10$. Using the tables you find a z-score of 1.28.

Calculator:

$invnorm(.10, 0, 1) = 1.28$

[STAT][TESTS][A:1-PROPZINTERVAL]...X, n, C.I.

Let's find the z-scores for the most common confidence levels!

Z-score for a 90% interval:

Z-score for a 95% interval:

Z-score for a 99% interval:

Choosing the Sample Size

In planning a study, we may want to choose a sample size that allows us to estimate a population proportion within a given margin of error.

The margin of error (ME) in the confidence interval for p is:

$$M.E. = z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Note: If $\hat{p}$ is unknown, then use $\hat{p} = 0.50$ for a conservative estimate.

Sampling Distribution of a Difference Between 2 Proportions

When our goal is to estimate a difference between two population proportions (or the size of a treatment effect), we select two independent random samples and use the difference in sample proportions as an estimate.

The **three conditions** when calculating confidence intervals for the difference between two proportions are:

1. **Random samples:** The observations represent a random sample of the population.
2. **Independence:** The samples are independently selected. Check $10n$.
3. **Sample size:** $n_1 p_1 \geq 10$ and $n_2 p_2 \geq 10$

$$n_1(1 - p_1) \geq 10 \text{ and } n_2(1 - p_2) \geq 10$$

Mean: $\hat{p}_1 - \hat{p}_2$ Standard Deviation: $\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$

C.I.: $(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$

Calculator: [STAT][TESTS][B:2-PROPZINTERVAL]...X, n, C.I.

Practice

1. Suppose we want to estimate the proportion of college students who have a high level of stress. A sample of 253 college students was collected and found that 56 of them had high levels of stress. Previously, we found that the estimated standard deviation for the sample proportion is 0.0261.

Calculate the margin of error that would be used to create a 95% confidence interval for the population proportion of students who had a high level of stress. *Round to 4 decimal places.*

2. We can now create our confidence interval for the proportion of college students who have a high level of stress. A sample of 253 college students was collected and found that 56 of them had high levels of stress. We found the margin of error to be 0.0512. Round to 4 decimal places.
 - (a) Find the lower bound of the confidence interval.
 - (b) Find the upper bound of the confidence interval.
3. The Gallup Youth Survey asked a random sample of 439 U.S. teens aged 13 to 17 whether they thought young people should wait to have sex until marriage. Of the sample, 246 said "Yes." Construct and interpret a 95% confidence interval for the proportion of all teens who would say "Yes" if asked this question.
 - a) Check conditions.
 - b) Find the z value.
 - c) Find $\hat{p}$.
 - d) Calculate the confidence interval.
 - e) Conclude (in context).

4. A company has received complaints about its customer service. The managers intend to hire a consultant to carry out a survey of customers. Before contacting the consultant, the company president wants some idea of the sample size that they will be required to pay for. One critical question is the degree of satisfaction with the company's customer service, measured on a 5-point scale. The president wants to estimate the proportion p of customers who are satisfied (that is, who choose either "satisfied" or "very satisfied," the 2 highest levels on the 5-point scale).

The president wants the estimate to be within 3% (.03) at a 95% confidence level. How large a sample is needed?

5. As part of the Pew Internet and American Life Project, researchers conducted two surveys in late 2009. The first survey asked a random sample of 800 U.S. teens about their use of social media and the Internet. A second survey posed similar questions to a random sample of 2253 U.S. adults. In these two studies, 73% of teens and 47% of adults said that they use social-networking sites. Use these results to construct and interpret a 95% confidence interval for the difference between the proportion of all U.S. teens and adults who use social-networking sites.

- a) Define parameters.
- b) Check conditions.
- c) Construct a 95% C.I.

d) Conclusion.